

August 31, 2026

Hexagon Autonomous Solutions

Dear Hiring Team,

I am writing to apply for the Software Engineer position with Hexagon Autonomous Solutions in Calgary. I recently completed my M.Sc. in Computer Science at the University of Calgary, where my work focused on geospatial data systems, C++/Python software, numerical methods, machine learning, and scalable data-processing pipelines. The opportunity to contribute to Hexagon's Algorithms group is especially interesting to me because it connects software engineering with positioning, data analysis, and demanding real-world applications.

My graduate research involved building software for large Earth observation datasets. I developed C++ and Python pipelines for processing, storing, retrieving, and analyzing satellite imagery, including a DGGS-based framework for multiresolution spatiotemporal data management. This work required careful algorithm design, attention to performance and scalability, and validation on large Canada-wide datasets. I also implemented performance-sensitive C++ components and improved tensor encoding with multithreading, reducing 10-tensor processing time from 233 seconds to 26 seconds.

I also worked on a digital elevation model super-resolution project that strengthened my experience with numerical evaluation, scientific data analysis, and automated experimentation. I built Python workflows using NumPy, pandas, Rasterio, Google Earth Engine, and PyTorch, then evaluated model outputs using RMSE, slope, TRI, TPI, and wavelet-based metrics. A major lesson from that project was that automated results are only useful when the evaluation method is trustworthy, which aligns closely with Hexagon's need for tools that analyze regression results, identify changes, and support tuning decisions.

Beyond research, I bring backend software experience from Asr Gooyesh Pardaz, where I developed Django REST APIs backed by PostgreSQL for NLP, text-to-speech, and speech-to-text products. I worked on authentication, payment-service integration, relational data models, and API behavior used by production services. Across both industry and research roles, I have used Git, Linux, Docker, Python, C++, and structured debugging practices to build maintainable systems.

Although my direct background is in geospatial data infrastructure and scientific computing rather than GNSS error modeling specifically, I am comfortable learning specialized technical domains through first principles, careful experimentation, and close collaboration with domain experts. My experience with spatial data, multiresolution representations, numerical evaluation, and performance-aware C++/Python development gives me a strong foundation for contributing to positioning-algorithm development and regression-analysis automation.

Thank you for considering my application. I would welcome the opportunity to discuss how my geospatial software, C++/Python, data-analysis, and research-engineering background could contribute to Hexagon Autonomous Solutions and the NovAtel Algorithms group in Calgary.

Sincerely,

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